

Arun Sabaratnam

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Technical Skills

Languages: Python, Java, JavaScript, SQL, MSSQL, C++, C#, Dart, Swift, HTML/CSS

Frameworks/Libraries: Spring Boot, Node.js, FastAPI, PySpark, Flask, React, Pandas, NumPy, Pytest

Cloud/DevOps: GCP, Azure, AWS, Docker, Kubernetes, CI/CD, Microsoft Fabric, MLFlow, Firebase

Tools: Git, Bitbucket, Jira, Linux (WSL), Postman, Redis, Flutter, Google Maps API

Experience

Scotiabank

Jan 2026 - Present

Software Engineer Intern | Python, Spring Boot, GCP, SQL

Toronto, ON

- Data Platforms Team

Statistics Canada

May 2025 - Sep 2025

AI Software Developer Intern | Python, Azure, Apache Spark, Microsoft Fabric

Ottawa, ON

- Modernized legacy Python workflows by engineering high-performance **PySpark** pipelines on **Microsoft Fabric**; optimized processing via strategic **shuffling**, **partitioning**, and **caching**.
- Engineered a **FastText** NLP model for business categorization, boosting accuracy by **20%** and leveraging **MLFlow** for robust model versioning and registry management.
- Created automated data pipelines to ingest and consolidate CRA tax records, facilitating seamless downstream analytics and reporting within the **MS Fabric** ecosystem.

Department of National Defence

Sep 2024 - Jan 2025

Full-Stack Developer Intern | Python, SQL, Docker

Ottawa, ON

- Optimized critical forecasting tools by refactoring database schemas with **Alembic** and **SQLAlchemy**, managing seamless migrations to reduce query latency by **30%**.
- Redesigned data-intensive UIs in **Dash** based on **user feedback sessions**, achieving a **30%** improvement in page load speeds and overall administrative workflow efficiency.
- Fortified system reliability by attaining **95%+ code coverage** with **Pytest** and modernizing **CI/CD** pipelines, realizing a **40%** reduction in production-level defects.

MindBridge AI

Jan 2024 - May 2024

Data Analyst Intern | Python, Java, Excel

Ottawa, ON

- Engineered data processing scripts using **Pandas** and **NumPy** to automate the ingestion of **50+ weekly financial datasets**, reducing client data onboarding wait times by **3x**.
- Refactored **Java Spring** backend APIs to eliminate performance bottlenecks, increasing system throughput and achieving a **15%** improvement in response times.

Payments Canada

May 2023 - Sep 2023

Software Developer Intern | Python, GCP, Google Apps Script

Ottawa, ON

- Engineered a serverless **Google Cloud Function** to automate employee probation updates via the **Dayforce API**, implementing **OAuth 2.0** authentication and error handling to reduce process time from **45 mins** to **2 mins**.
- Developed **Google Apps Scripts** to fetch external API data directly into Sheets and automated the renaming of **300+ files**, ensuring **100% compliance** with data retention policies.
- Authored technical documentation and presented the automation suite to leadership, ensuring seamless team adoption and long-term maintainability.

Projects

Ford EV Tracker | Flutter, Dart, Python, Flask, Firebase

Jan 2024

- Developed a **Flutter (Dart)** mobile application for electric vehicles, utilizing a **Flask** backend and **REST** endpoints to automate real-time route optimization with **Google Maps API** and **Firebase** for user data synchronization.

Education

University of Ottawa

Expected May 2026

Bachelor of Applied Science in Software Engineering

Ottawa, ON